

EE345/L: Linear Algebra and Machine Learning

Narayana Santhanam, Fall 2025

Welcome to EE345/L!

The course is divided into a sequence of modules which you can find at

<https://uhm-descartes.github.io/ee345>

You will submit assignments on Lamaku, but the above site is the primary content site.

AI in the loop This course has an associated AI companion that you will have access to. We will be using this extensively to accelerate learning, particularly for certain parts of the course. As I will tell you repeatedly, if you are doing only a subset of what chatGPT can do, you really should be worried—a subscription to chatGPT pro is only \$240, and anyone hiring you will wonder if your \$100k salary is worth it when chatGPT can do what it does for \$240.

The emphasis of this course is to go beyond. Of course, as humans, you need to learn some basics that form the bedrock of your thinking process. Whether or not chatGPT can do those (like calculus, basic lines of thinking in this course) is irrelevant. But you should simultaneously learn to outgrow these processes. For this reason, we will try to use this AI companion to push beyond the syllabus.

Modules You will find all modules on the github site above. We will be covering a subset of the material in the modules. Each module is focused on a specific topic and is assigned a range of dates (which is subject to change). Each module usually has

- Readings, which include the part of the text that you can use to prepare, follow along, or review, as well as handouts. Notice that we do not necessarily follow the same order of the text, sometimes deemphasizing some material in the text, and sometimes adding in additional material.
- Lectures in class: these are not optional. See next part.
- Experiences primarily include problems we solve in class in groups, with my help and with the AI companion. There is a portion of the grade assigned to this team-solving. Skipping classes will hurt your grade in this section.
- Quizzes—these are released on Lamaku at the beginning of the module. The quizzes are only due towards the end of the module. But they are provided at the beginning of the module so you can anticipate some highlights of the material to come.
- Final project: we will have only one midterm. In lieu of another midterm, we will have a project. I will announce the topic at the end of September.

HW You will turn in the solutions for the in-class problems in each module, and they will be graded. This portion is a sanity check to ensure that you understood the material.

Text The text we will use is *Introduction to linear algebra* by Gilbert Strang. The edition really does not matter too much, anything from the 4th edition is fine.

Exams There will be one midterm, a project and a final exam.

Grading This is tentative, and will be updated on how the class experience goes. The breakdown of credit is as follows: quizzes (15%), in-class problems (15%), project (15%), midterm (10%), final (25%). The remaining 20% is for the labs.

Office hours Office hours TBD. If you wish to see me any other time (on MWF), please email me.